

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026 — ANSWER KEY (Version 1)

MATHEMATICS — Class IX

**Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation |
Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry**

Total Marks: 50

Time Allowed: 2 Hours 15 Minutes

Version: 1

SECTION A — MCQ Answer Key ($10 \times 1 = 10$ Marks)

#	Answer	Explanation
1	B	$(3^4 \times 3^{-2}) \div 3^{-3} = 3^{4-2} \div 3^{-3} = 3^2 \div 3^{-3} = 3^{2-(-3)} = 3^5$. 1 mark SLO: Apply the laws of exponents to simplify expressions with real exponents. Source in her textbook: Unit 1, §1.5.3 Properties of Exponents — product rule ($a^m \cdot a^n = a^{m+n}$) and quotient rule ($a^m \div a^n = a^{m-n}$).
2	C	$\sqrt{48} = \sqrt{(16 \times 3)} = 4\sqrt{3}$. $\sqrt{27} = \sqrt{(9 \times 3)} = 3\sqrt{3}$. So $\sqrt{48} + \sqrt{27} = 4\sqrt{3} + 3\sqrt{3} = 7\sqrt{3}$. 1 mark SLO: Apply the product rule for radicals to simplify and combine radical expressions. Source in her textbook: Unit 1, §1.4.3 Properties of Radicals — Product Rule for Radicals, Example 6. <i>Watch out: $\sqrt{48} + \sqrt{27}$ is NOT $\sqrt{75}$ — you can never add the numbers under two different radicals directly. Always simplify each radical to its simplest form first (break 48 into 16×3 and 27 into 9×3), and only THEN add the like radical terms: $4\sqrt{3} + 3\sqrt{3} = 7\sqrt{3}$.</i>
3	A	Let $\log_5(1/25) = y$. Then $5^y = 1/25 = 5^{-2}$, so $y = -2$. 1 mark SLO: Define logarithm of a number to the base a . Source in her textbook: Unit 2, §2.2 Definition of Logarithm ($\log_a b = y \Leftrightarrow a^y = b$).
4	D	$128 = 2^7$, so $\log 128 = 7 \times \log 2 = 7 \times 0.3010 = 2.1070$. 1 mark SLO: Apply the laws of logarithm (power law) to convert exponentiation into multiplication and evaluate. Source in her textbook: Unit 2, §2.4 Laws of Logarithm — power law, $\log(a^n) = n \cdot \log a$; Exercise 2.2-style "evaluate using given log values".
5	B	$x^2 - 14x + 49$: need two numbers multiplying to 49 and adding to $-14 \Rightarrow -7$ and -7 . So $x^2 - 14x + 49 = (x-7)^2$. Check: $(x-7)^2 = x^2 - 14x + 49$. ✓ 1 mark SLO: Factorize expressions of the type $a^2 - 2ab + b^2$ (perfect square trinomial). Source in her textbook: Unit 4, §4.1 Factorization Types — $a^2 \pm 2ab + b^2 = (a \pm b)^2$.
6	C	$15a^3b^2 = 3 \times 5 \times a^3 \times b^2$. $20a^2b^3 = 2^2 \times 5 \times a^2 \times b^3$. Common numeric factor = 5; lowest power of $a = a^2$; lowest power of $b = b^2$. HCF = $5a^2b^2$. 1 mark SLO: Find highest common factor (HCF) of algebraic expressions using the factor method. Source in her textbook: Unit 4, §4.2 HCF by Factorization (monomial HCF using lowest common powers).
7	A	$ -8 = 8$ (distance of -8 from zero) and $ 3 = 3$. So $ -8 + 3 = 8 + 3 = 11$. 1 mark SLO: Define absolute value. Source in her textbook: Unit 5, §5.3.1 Absolute Value. <i>Watch out: the bars in -8 are NOT brackets — they mean "distance from zero," which is never negative. The book states this directly in §5.3.1: "$5 = 5$ and $-5 = 5$." So $-8 = 8$, not -8. Once both bars are turned into their positive distances, the rest is ordinary addition.</i>
8	D	$-3 \leq 2x - 1 \leq 5$. Add 1 to <u>all three parts</u> : $-2 \leq 2x \leq 6$. Divide <u>all three parts</u> by 2: $-1 \leq x \leq 3$, i.e. $x \in [-1, 3]$. 1 mark SLO: Solve linear inequalities with rational coefficients. Source in her textbook: Unit 5, §5.4 Compound Inequalities (operating on all three parts of a "sandwiched" inequality together).

		<i>Watch out: whatever you do (add, subtract, divide), you must do it to BOTH outer bounds AND the middle — not just the middle and one side. Adding 1 to -3 gives -2 (not 1), so after dividing by 2 the left bound is -1, not dropped altogether. The final interval keeps its negative left edge: [-1, 3].</i>
9	B	$d = \sqrt{[(9-0)^2 + (12-0)^2]} = \sqrt{(81+144)} = \sqrt{225} = 15.$ 1 mark SLO: Find distance between two points in the plane. Source in her textbook: Unit 7, §7.1 The Distance Formula, Example 2.
10	C	Mid-point = $((-5+1)/2, (3+(-7))/2) = (-4/2, -4/2) = (-2, -2).$ 1 mark SLO: Derive the mid-point formula and calculate the mid-point of a line segment. Source in her textbook: Unit 7, §7.2 Mid-Point of Two Given Points, Example 5. <i>Watch out: the mid-point formula AVERAGES the coordinates — it divides the sum by 2, it does not just add them. $(-5+1)/2 = -4/2 = -2$, not -4. Do the division every time, on both the x-sum and the y-sum.</i>

SECTION B — Short Question Answers (6 × 4 = 24 Marks)

2(i) Using the laws of exponents, simplify: **4 marks**

(a) $16^{3/4} \times 16^{-1/4} = 16^{(3/4 - 1/4)} = 16^{1/2}$ **1 mark**

$16^{1/2} = \sqrt{16} = 4$ **1 mark**

(b) $(x^{-2}y^3)^{-2} \div (x^3y^{-1})^2 = x^4y^{-6} \div x^6y^{-2}$ **1 mark**

$= x^{(4-6)}y^{(-6-(-2))} = x^{-2}y^{-4} = 1/(x^2y^4)$ **1 mark**

2(ii) 4 marks

(a) $\log[(p^3\sqrt{q})/r^4] = \log(p^3) + \log(q^{1/2}) - \log(r^4) = 3 \log p + \frac{1}{2} \log q - 4 \log r$ **2 marks**

(b) $\log 0.9 = \log(9/10) = \log 9 - \log 10 = 2 \log 3 - 1$ **1 mark**

$= 2(0.4771) - 1 = 0.9542 - 1 = -0.0458$ **1 mark**

2(iii) Find the second polynomial Q, given $P = x^2+2x-24$, $HCF = x-4$, $LCM = x^3+x^2-26x+24$. **4 marks**

Factorize P: $x^2+2x-24 = (x+6)(x-4)$ **1 mark**

Factorize LCM: $x^3+x^2-26x+24 = (x-1)(x+6)(x-4)$ **1 mark**

Using $HCF \times LCM = P \times Q$: $Q = (HCF \times LCM) \div P = [(x-4) \times (x-1)(x+6)(x-4)] \div [(x+6)(x-4)]$ **1 mark**

$Q = (x-4)(x-1) = x^2 - 5x + 4$ **1 mark**

2(iv) Simplify to lowest form: $1/(x^2-1) + 1/(x^2-3x+2) + 1/(x^2+3x+2)$ **4 marks**

Factorize each denominator: $x^2-1=(x-1)(x+1)$; $x^2-3x+2=(x-1)(x-2)$; $x^2+3x+2=(x+1)(x+2)$ **1 mark**

LCM of denominators = $(x-1)(x+1)(x-2)(x+2)$ **1 mark**

Numerator = $(x-2)(x+2) + (x+1)(x+2) + (x-1)(x-2) = (x^2-4) + (x^2+3x+2) + (x^2-3x+2) = 3x^2$ **1 mark**

Sum = $3x^2 / [(x^2-1)(x^2-4)]$ **1 mark**

2(v) 4 marks

(a) $|2x+1| = 13 \Rightarrow 2x+1 = 13$ or $2x+1 = -13$ **1 mark**

Case 1: $2x = 12 \Rightarrow x = 6$. Case 2: $2x = -14 \Rightarrow x = -7$. Verify: $|2(6)+1| = |13| = 13 \checkmark$. $|2(-7)+1| = |-13| = 13 \checkmark$. Both solutions check out: **x = 6, x = -7** **1 mark**

(b) $-7 < 3x+2 \leq 14 \Rightarrow$ subtract 2 from all parts: $-9 < 3x \leq 12$ **1 mark**

Divide all parts by 3: $-3 < x \leq 4$, i.e. solution set = $\{x \mid -3 < x \leq 4, x \in \mathbb{R}\}$ **1 mark**

2(vi) Show that A(5,7), B(2,3), C(10,-3) are vertices of a right-angled triangle. 4 marks

$$AB = \sqrt{[(5-2)^2 + (7-3)^2]} = \sqrt{(9+16)} = \sqrt{25} = 5 \quad 1 \text{ mark}$$

$$BC = \sqrt{[(10-2)^2 + (-3-3)^2]} = \sqrt{(64+36)} = \sqrt{100} = 10 \quad 1 \text{ mark}$$

$$AC = \sqrt{[(10-5)^2 + (-3-7)^2]} = \sqrt{(25+100)} = \sqrt{125} \quad 1 \text{ mark}$$

$$AB^2 + BC^2 = 25 + 100 = 125 = AC^2 \Rightarrow \text{by the converse of Pythagoras' theorem, } \triangle ABC \text{ has a right angle at B.} \quad 1 \text{ mark}$$

SECTION C — Long Question Answers (2 × 8 = 16 Marks)

3(i)(a) Reduce to lowest form: $[(x^2-25)/(x^2+8x+15)] \times [(x^2+3x)/(x^2-3x-10)]$ 4 marks

Factorize all four expressions: $x^2-25=(x-5)(x+5)$; $x^2+8x+15=(x+3)(x+5)$; $x^2+3x=x(x+3)$; $x^2-3x-10=(x-5)(x+2)$ 2 marks

$$\text{Product} = [(x-5)(x+5) \times x(x+3)] / [(x+3)(x+5) \times (x-5)(x+2)] \quad 1 \text{ mark}$$

Cancel common factors (x-5), (x+5), (x+3): result = $x/(x+2)$ 1 mark

3(i)(b) Find the square root of $9x^4-12x^3+10x^2-4x+1$ by the division method. 4 marks

Step 1: $\sqrt{(9x^4)} = 3x^2$. Subtract $(3x^2)^2=9x^4$: remainder = $-12x^3+10x^2-4x+1$ 1 mark

Step 2: divide $-12x^3$ by $2(3x^2)=6x^2 \Rightarrow -2x$. New quotient: $3x^2-2x$. $(6x^2-2x)(-2x) = -12x^3+4x^2$. New remainder = $6x^2-4x+1$ 1 mark

Step 3: divide $6x^2$ by $2(3x^2-2x)=6x^2-4x \Rightarrow +1$. New quotient: $3x^2-2x+1$. $(6x^2-4x+1)(1) = 6x^2-4x+1$. Remainder = 0 1 mark

Since the remainder is 0, the square root is $\pm(3x^2-2x+1)$. 1 mark

3(ii) Survey pillars P(1,2), Q(4,6), R(7,10). 8 marks

(a) Using $d = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$: 1 mark

$$PQ = \sqrt{[(4-1)^2 + (6-2)^2]} = \sqrt{(9+16)} = \sqrt{25} = 5 \quad 1 \text{ mark}$$

$$QR = \sqrt{[(7-4)^2 + (10-6)^2]} = \sqrt{(9+16)} = \sqrt{25} = 5 \quad 1 \text{ mark}$$

$$PR = \sqrt{[(7-1)^2 + (10-2)^2]} = \sqrt{(36+64)} = \sqrt{100} = 10 \quad 1 \text{ mark}$$

(b) $PQ + QR = 5 + 5 = 10 = PR$ 1 mark

Since $d(P,Q) + d(Q,R) = d(P,R)$, the three points lie on the same straight line $\Rightarrow$ P, Q, R are collinear. 1 mark

(c) Mid-point of PR = $((1+7)/2, (2+10)/2) = (8/2, 12/2) = (4, 6)$ 1 mark

This is exactly the point Q(4, 6) — so Q is the mid-point of PR, confirming the pillars are evenly spaced along the canal. 1 mark

Checked by: _____

Total Score: _____

Chapters Covered: Unit 1 (Real Numbers) | Unit 2 (Logarithms) | Unit 4 (Factorization & Algebraic Manipulation) | Unit 5 (Linear Equations & Inequalities) | Unit 7 (Coordinate Geometry)

— End of Answer Key —